

Aarav Babu

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EDUCATION

University of California, Riverside

MSc in Computer Science, GPA: 3.97

Riverside, CA

Expected March 2026

PES University

BTech in Computer Science and Engineering

Bangalore, India

June 2024

WORK EXPERIENCE

Full Stack Software Engineer Intern

Dyrt Labs (SaaS sustainability startup)

July 2025 – Present

Los Angeles, CA

- Built an end-to-end AI Waste Bills system using React, TypeScript, Node.js, Python, AWS OCR, AWS S3, and OpenAI to digitize invoices with categorization using RAG few-shot examples, reducing manual processing by 80%.
- Owned auto-invoicing billing backed by PostgreSQL, delivering REST API integrations with QuickBooks to automate invoice creation and payment workflows.
- Shipped 5+ production features across billing and operations by building reusable React + TypeScript components and integrating Supabase (PostgreSQL), internal APIs, and data-driven UIs in Agile sprints.
- Improved CI/CD reliability in GitHub Actions via automated checks and an AI bugbot, accelerating delivery through AI-assisted coding and code reviews using Cursor and Claude Code.
- Hardened releases with JUnit, Cypress, and Playwright tests, and implemented FullStory analytics to understand user flows and reduce UX friction.
- Mentored junior interns through pipeline setup, debugging, and onboarding for local dev environments and deployment workflows.

Software Engineering Intern

GSK (GlaxoSmithKline)

January 2024 – July 2024

Bangalore, India

- Built an Azure-hosted ERP automation tool using Python, SQL, and REST APIs to validate and triage security tickets, reducing manual processing time and improving data quality.
- Integrated APIs across SAP and ServiceNow workflows and developed a proof-of-concept ML ticket classifier to improve routing accuracy.

Data Analytics and Machine Learning Intern

Caterpillar Inc.

June 2023 – August 2023

Bangalore, India

- Developed ML models for reason-code classification using Python, pandas, NumPy, and PyTorch, achieving 82% accuracy.
- Built 5+ Power BI dashboards for automation and operations metrics and automated barcode-scanning plus data-retrieval modules to improve consistency.

Research Intern

Center for Data Science and Applied Machine Learning, PES University

January 2022 – January 2023

India

- Researched automatic annotation methods for NLP sentiment analysis on code-mixed e-commerce reviews, benchmarking against manual labeling.
- Built a 4,000+ review dataset across 80+ products using Python web scraping and structured data annotation; validated ML baselines (87.8% English, 80.9% code-mixed) and contributed to a peer-reviewed paper.

PROJECTS

AI Driven Email Chatbot for Law Enforcement | React, TypeScript, Python, SQL, Flask, GCP, Docker, Twilio, LLMs

- Delivered Craigslist SafeBot for the Riverside Police Department, automating 90% of workflows via an operator web UI and REST APIs for LLM email generation, risk scoring, and retrieval pipelines.

Automated Valuation of Used Items | Python, scikit-learn, Web-Scraping, Streamlit, Selenium, MySQL

- Ingested 50K+ listings/day; trained Random Forest models (93%) and shipped a Streamlit UI backed by MySQL.

AI Powered Scam Call Detection | Python, PyTorch, TensorFlow, Flask, Twilio API

- Built a CNN + BiLSTM scam detector (92%) served via Flask APIs integrated with Twilio for real-time call handling.

TECHNICAL SKILLS

Languages: JavaScript, TypeScript, Python, SQL, C++, Java (coursework)

Frontend: React, HTML, CSS, Responsive Design, Component-Based UI

Backend and Data: Node.js (Express), REST APIs, GraphQL, Supabase (PostgreSQL)

Cloud and Tools: AWS, Azure, GCP, Docker, Git, GitHub Actions, CI/CD, Cypress